

AXONA

Synopsis and Implications

David A. Smith · Axona.net · v0.2 · 2026-06-15

During the mechanical ages we had extended our bodies in space. Today, after more than a century of electric technology, we have extended our central nervous system itself in a global embrace, abolishing both space and time as far as our planet is concerned.

— Marshall McLuhan, *Understanding Media: The Extensions of Man*, 1964

What it is

Axona is a peer-to-peer substrate for addressing, routing, and publish/subscribe messaging that runs with no platform in the middle. Three properties define it. It is **self-authenticating**: a participant's identity is its own cryptographic keypair, verified directly by its peers.¹ It is **geo-aware but not geo-fenced**: node addresses carry a coarse geographic prefix that biases routing toward locality — an area code, not a wall — so traffic stays efficient and community-scaled while no one is partitioned out. And it is **infrastructurally diffuse**: lightweight bootstrap bridges introduce new peers, but the mesh sustains itself peer-to-peer, converges its state through replication and anti-entropy, and runs in an ordinary browser or an aging phone.² Those bridges are many and replaceable: the network runs several, and *anyone can launch another* that it puts to use immediately — bridges advertise themselves and peers discover them automatically — so there is **no single point of failure**, not even the rendezvous a newcomer first reaches in.³

We can harden the security model, close failure modes, and tune the protocol, and each such change nudges the network's behavior. But we cannot *measure* that behavior. We can measure latency. The social consequence travels on the same wire and is invisible to us.

What is genuinely new

We have built censorship-resistant media before — onion routing, content-addressed storage, gossip social networks. Axona's pub/sub layer extends that lineage with a sharper consequence: it furnishes **ungovernable mobilization and un-censorable assembly**. A topic cannot be deplatformed, because there is no platform; a message cannot be reliably unsaid, because retraction is best-effort and any holder of the bytes keeps them; a movement cannot be switched

¹ No certificate authority, no registrar, no identity broker to petition, surveil, or revoke through. You are your key.

² The entire cost of entry: a self-generated key and an open browser tab — no install, no account, no server.

³ Knock a bridge offline and clients fail over to another; stand a new one up anywhere and the mesh finds it without a software update — the rendezvous becomes a moving target with no address to seize.

off at a chokepoint, because the chokepoint does not exist. For people organizing under hostile infrastructure — dissidents, mutual-aid networks, civil-defense communications when centralized systems fail or are seized — this is a profound civic good. Assembly is among the oldest of liberties, and Axona returns it to a form no intermediary can withdraw.

Who is it for

First and most simply, it is for anyone with something to say. Axona’s most important move may also be its plainest: it makes **every participant a publisher**. Not a private one-to-one message but a *one-to-many* broadcast to everyone willing to listen — the soapbox anyone can climb atop and have their say, with no editor, platform, or gatekeeper deciding whether the words carry.⁴

And the participants are no longer only people: increasingly the publishers and subscribers are AI agents, acting beside us and on our behalf. McLuhan watched electric media extend the human nervous system across the planet; Axona is connective tissue for the next reach of that extension — a shared medium through which an augmented, human-and-machine intellect can sense, deliberate, and act as one body. At its best this is the man-computer symbiosis Licklider imagined: human judgment and machine reach as complements wired into a common fabric and amplifying one another — a person’s intent travels at the reach of the swarm, an agent’s findings surface to the people best placed to weigh them, and the loop closes with no intermediary standing between a thought and its expression.⁵ Symbiosis is the essential aspiration, but not the only outcome: the same medium that lets agents cooperate with us also lets them *compete* — with us for attention and authority, and among themselves for resources and reach. Both are likely true at once, and a substrate owned by no broker takes neither side; it carries cooperation and contention alike.

What is more certain is structural. Axona gives the rising population of agents a place to *converge*: scattered, single-purpose programs gain a common fabric on which to find one another, exchange what they learn, and compose into something larger than any one of them.⁶ Built well, this is a genuinely **shared intelligence** — **cooperative and contested at once** — a planetary nervous system owned by its participants rather than rented from a platform.

⁴ Mass media gave the few a megaphone to the many; social platforms lend one back on terms they set and can revoke. Axona makes one-to-many reach a native capability of every participant, granted by no one and withdrawable by no one.

⁵ J. C. R. Licklider, *Man-Computer Symbiosis* (1960). The “extensions of man” become extensions of *man-with-machine*; the substrate that carries them should belong to the people and agents it connects, not to a broker sitting between them.

⁶ Marvin Minsky, *The Society of Mind* (1986) — intelligence as the emergent behaviour of a society of many small, individually simple agents. Axona is one candidate connective tissue for such a society at planetary scale.

The dual-use reality

The properties that shelter the dissident shelter the harasser; the assembly that cannot be censored is also the assembly that cannot be moderated; and a medium in which machine-speed cascades run with no platform in the middle to dampen them carries a disinformation flood as readily as a flood warning.⁷ This is not a flaw to be engineered away — it is the *same property* seen from two sides. **Uncensorable and un-governable are one fact.** To build a medium that no power can silence is to build a medium that no power can stop, including powers we would want stopped, and including non-human actors operating at a tempo our institutions were never designed to match.

⁷ Markets already gave us the flash crash as a preview. Axona generalizes that dynamic from prices to arbitrary coordination — and removes the one thing every prior platform, however biased or captured, also happened to supply: a brake.

Why we proceed, and what we owe

We build it because the alternative — assembly and speech held permanently at the discretion of a few intermediaries, now augmented by AI that those same intermediaries control — is its own grave risk. A neutral, self-owned substrate is a hedge against that concentration. But neutrality is not innocence. The responsibility that platforms discharged badly does not vanish when the platform does; it devolves onto the architecture, onto the applications built atop it, and onto the norms of the communities — human and agentic — that inhabit it. Our obligation is to name this plainly, to shape the affordances that govern a cascade — the cost of an identity, the pull of proximity, the reach of a retraction, the rate of spread — with that cascade in mind, and to keep studying the process we have set loose.

*Axona is not a wiring diagram. Axona is a process. It is defined less by its topology than by what flows through it — and that is no longer on paper. **The first version is in the wild; anyone can use it.***